

Prashant JAYANNAVAR

PhD | University of Illinois Urbana-Champaign

@ paj3@illinois.edu 🌐 prashant-jayan21.github.io 🎓 Google Scholar
🐙 github.com/prashant-jayan21 🔗 linkedin.com/in/prashantjayannavar 🐦 @p_jayannavar
📍 Thomas M. Siebel Center for Computer Science, 201 North Goodwin Avenue, Urbana, IL 61801

I am a recent CS PhD graduate from UIUC, advised by Prof. Julia Hockenmaier, specializing in **Natural Language Processing (NLP)**. My research centers on instruction-following agents in **embodied dialogue**, using games as simulated environments. Themes include human-AI collaboration, spatial reasoning, SLMs, synthetic data, and evaluation benchmarks.

Previously, I spent 2.5 years as an **Applied Scientist** at the AI startup x.ai (acquired by Bizzabo), building a task-oriented dialogue agent to automate email-based meeting scheduling.

I am actively seeking **Research Scientist** or **Applied Scientist** roles.

EDUCATION

- | | |
|-------------------------|---|
| Sept 2017
- Dec 2025 | PhD in Computer Science , University of Illinois at Urbana-Champaign, <i>Urbana, IL</i>
Advisor: Prof. Julia Hockenmaier (UIUC)
Committee: Prof. Dilek Hakkani-Tur (UIUC), Prof. ChengXiang Zhai (UIUC), Prof. Massimo Poesio (QMUL, UK)
Research Areas: Grounded Instruction Following, Task-Oriented Dialogue, Embodied AI, Datasets and Evaluation, Synthetic Data, Spatial Reasoning in LLMs |
| Jan 2015 | MS in Computer Science (Track: Machine Learning) , Columbia University, <i>New York, NY</i> |
| May 2013 | B.Tech in Computer Science and Engineering , National Institute of Technology Rourkela, <i>Rourkela, India</i> |

RESEARCH & INDUSTRY EXPERIENCE

- | | |
|-----------------------|---|
| Dec 2025
Sept 2017 | Research Assistant, UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN, Urbana, IL
Advisor: Prof. Julia Hockenmaier
My research focuses on instruction-following agents and spatial intelligence in embodied dialogue. It develops a novel research ecosystem for this domain by using games as simulated environments:
Part 1: A Novel Testbed <ul style="list-style-type: none">➤ Created the Minecraft Dialogue Corpus (MDC), a pioneering use of Minecraft in this domain.➤ Built and open-sourced the data collection platform.➤ Formulated the Builder Action Prediction (BAP) task for instruction following: evaluation, baselines (CNN+GRU), and data augmentation methods. Part 2: BAP v2: An Enhanced Task Framework <ul style="list-style-type: none">➤ Developed an improved evaluation benchmark for BAP.➤ Designed novel dialogue simulators to generate rich synthetic embodied dialogues.➤ Designed a Curriculum Learning strategy to leverage this data and boost spatial reasoning in models.➤ Established newer baseline architectures using Transformers and early LLMs (BERT). Part 3: Into the Age of Modern LLMs <ul style="list-style-type: none">➤ Developed the new state-of-the-art model, Llama-CRAFTS, achieving F1 of 53.0, +6.0 over prior SOTA (post-training Llama-3-8B with SFT/instruction tuning).➤ Demonstrated that BAP v2 provides a key measure of spatial capabilities and brittleness of LLMs. <div><div>Grounded Instruction Following</div><div>Embodied AI</div><div>Spatial Reasoning</div><div>Task-Oriented Dialogue</div><div>LLMs</div><div>Synthetic Data</div><div>Evaluation Benchmarks</div><div>Data Collection</div><div>Minecraft</div></div> |
| June 2017
Feb 2015 | Applied Scientist, x.AI (ACQUIRED BY BIZZABO), New York, NY
Built a production NLP pipeline for temporal information extraction from unstructured email text. <ul style="list-style-type: none">➤ Developed temporal expression extraction and normalization, combining ML with syntactic parsing and a proprietary temporal markup language.➤ Built a dialog act classifier to interpret user intent behind temporal expressions.➤ Deployed microservices integrating these capabilities into the main product.➤ Implemented online and offline metrics to benchmark and monitor model performance.➤ Authored annotation guidelines and supervised annotation, collaborating with annotators, engineers, UI designers, and product leaders. <div><div>Temporal Information Extraction</div><div>Dialogue Systems</div><div>Data Annotation</div><div>NLP Engineering</div><div>Microservices</div></div> |

Dec 2014	Research Assistant, COLUMBIA UNIVERSITY, New York, NY
Jun 2014	Advisor: Prof. Owen Rambow Demonstrated the utility of NLP in digital humanities research. ➤ Enhanced the SINNET system for social network extraction and applied it to nineteenth-century British novels to test long-standing literary theories. ➤ Validated the system by showing high correlation with human annotations and cross-genre generalization from news articles (training data) to literary texts. ➤ Challenged the prevailing interpretation of the theories, arguing that prior quantitative findings support rather than disprove them, and generated stronger evidence for this revised interpretation.
	NLP in Digital Humanities Social Network Analysis Computational Literary Analysis

PUBLICATIONS

J=JOURNAL, C=CONFERENCE, W=WORKSHOP, D=DEMO, R=REPORT

Selected Papers

- [J.1] **BAP v2: An Enhanced Task Framework for Instruction Following in Minecraft Dialogues** [[pre-MIT Press publication version](#) | [code](#) | [blog](#)]
 Prashant Jayannavar, Liliang Ren, Marisa Hudspeth, Risham Sidhu, Charlotte Lambert, Ariel Cordes, Elizabeth Kaplan, Anjali Narayan-Chen, Julia Hockenmaier
Computational Linguistics journal (*selected for oral presentation at ACL 2026) [Computational Linguistics; 2025]
- [C.1] **MDC-R: The Minecraft Dialogue Corpus with Reference** [[paper](#)]
 Chris Madge, Maris Camilleri, Paloma Carretero Garcia, Vanja Karan, Juexi Shao, Prashant Jayannavar, Julian Hough, Benjamin Roth, Massimo Poesio
Under review; arXiv preprint arXiv:2506.22062 [arXiv; 2025]
- [R.1] **Human in the Loop Learning through Grounded Interaction in Games (Dagstuhl Perspectives Workshop 24492)** [[paper](#) | [website](#)]
 Raffaella Bernardi, Julia Hockenmaier, Udo Kruschwitz, Prashant Jayannavar, Massimo Poesio
Dagstuhl Reports, Vol. 14, Issue 12, Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2025) [Dagstuhl Reports 2025]
- [C.2] **Learning to execute instructions in a Minecraft dialogue** [[paper](#) | [video](#) | [slides](#) | [code](#) | [website](#)]
 Prashant Jayannavar, Anjali Narayan-Chen, Julia Hockenmaier
58th Annual Meeting of the Association for Computational Linguistics (Virtual) [ACL 2020]
- [C.3] **Collaborative Dialogue in Minecraft** [[paper](#) | [data](#) | [code \(models\)](#) | [code \(platform\)](#) | [website](#)]
 Anjali Narayan-Chen*, Prashant Jayannavar*, Julia Hockenmaier (* = Equal Contribution)
57th Annual Meeting of the Association for Computational Linguistics, Florence, Italy [ACL 2019]

Other Papers

- [C.4] **Lara - Human-guided collaborative problem solver: Effective integration of learning, reasoning and communication** [[paper](#) | [supplemental](#) | [blog](#) | [video](#)]
 Harsha Kokel, Mayukh Das, Rakibul Islam, Julia Bonn, Jon Cai, Soham Dan, Anjali Narayan-Chen, Prashant Jayannavar, Janardhan Rao Doppa, Julia Hockenmaier, Sriraam Natarajan, Martha Palmer, Dan Roth
Tenth Annual Conference on Advances in Cognitive Systems, Arlington, Virginia [ACS 2022]
- [D.1] **Human-guided Collaborative Problem Solving: A Natural Language based Framework** [[paper](#) | [blog](#) | [video](#)]
 Harsha Kokel, Mayukh Das, Rakibul Islam, Julia Bonn, Jon Cai, Soham Dan, Anjali Narayan-Chen, Prashant Jayannavar, Janardhan Rao Doppa, Julia Hockenmaier, Sriraam Natarajan, Martha Palmer, Dan Roth
Thirty-First International Conference on Automated Planning and Scheduling, Guangzhou, China [ICAPS 2021]
- [W.1] **Validating Literary Theories Using Automatic Social Network Extraction** [[paper](#)]
 Prashant Jayannavar, Apoorv Agarwal, Melody Ju, Owen Rambow
NAACL HLT Fourth Workshop on Computational Linguistics for Literature, Denver, Colorado [CLFL, NAACL HLT 2015]

MENTIONS & MEDIA COVERAGE

- **Prof. Martha Palmer** (July 2023) — *My Big, Fat 50-Year Journey* (2023 ACL Lifetime Achievement Award acceptance speech). [[Transcript in Computational Linguistics Journal](#)]
- **Queen Mary University of London EECS News** (Nov 2021) — *EECS researcher wins over £1million of funding for research project* (featuring Prof. Massimo Poesio). [[Article](#)]
- **U.S. Army DEVCOM Army Research Laboratory Public Affairs** (July 2021) — *Army award-winning research to transform Soldier-robot communication*. [[Article](#)]

TECHNICAL SKILLS

Programming:	Python, Scala, Java, C++
ML and NLP Libraries:	PyTorch, NumPy, SciPy, scikit-learn, Stanford CoreNLP, NLTK
LLM Engineering:	Hugging Face (Transformers, Datasets, TRL, PEFT, Pipeline), Post-training (SFT), QLoRA, Distributed Training (Accelerate), BitsAndBytes
Data Analysis & Viz:	Pandas, Matplotlib, Plotly, Jupyter Widgets (ipywidgets)
Tools & Platforms:	Git, TensorBoard, Slurm, LaTeX, Unix/Linux, AWS, MongoDB, gRPC, Protobuf

MENTORSHIP

Master's students

- **Amit Athani** (MS CS, UIUC → Palantir) — Master's Thesis (2023).
Topic: *Introducing Builder Perspective in Minecraft Architect Utterance Generation*. [Thesis]
- **Dhruv Agarwal** (MS CS, UIUC → Amazon) — Master's Thesis (2019).
Topic: *Multi Task Learning and Incorporating Common Sense Knowledge for Question Answering*. [Thesis]

Undergraduate students

- **Marisa Hudspeth** (BS CS, Rhodes College → PhD, UMass, Amherst) — NSF DREU Research Internship (2021).
Project: *Synthetic Data Generation and Dialogue Simulation for Embodied AI in Minecraft*.
Outcome: Co-author on **publication [J.1]**. [DREU Program | Paper]
- **Charlotte Lambert** (BA CS, Vassar College → PhD CS, UIUC), **Ariel Cordes** (BA CS, UMN–Morris), **Elizabeth Kaplan** (BS CS, NCSU → Amazon) — NSF DREU Research Internship (2019).
Project: *Virtual World Context Encoding for Grounded Dialogue in Minecraft*.
Outcome: Received the **Best Research Project Award**; Co-author on **publication [J.1]**. [DREU Program | Report | Paper | Award Details]
- **Hetvi Patel** (BS CS, UIUC), **Sana Madhavan** (BS CS+Linguistics, UIUC) — UIUC CS STARS Research Program (2024).
Project: *Data Annotation for Fairer Evaluation for the Minecraft Builder Task*.
Outcome: Acknowledged in publication [J.1]. [CS STARS Program]

TEACHING EXPERIENCE

May 2025 Sept 2021	Teaching Assistant, UNIVERSITY OF ILLINOIS AT URBANA- CHAMPAIGN, Urbana, IL Performed TA duties including holding office hours and providing academic support for students in CS 447: Natural Language Processing (taught by Prof. Julia Hockenmaier).
May 2014 Feb 2014	Academic Tutor, COLUMBIA UNIVERSITY, New York, NY Tutored Columbia student athletes in Intro to CS and Programming in MATLAB (COMS W1005) and Calculus III (MATH V1201) as part of <i>Columbia's Student-Athlete Enrichment Services program</i> .